Smart Assisstive Stick With ML-Based Fall Detection

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***Abstract*—Falls are a major safety concern among elderly and visually impaired individuals, particularly for those who depend on walking aids for daily mobility [1]. Conventional walking sticks provide limited assistance beyond physical support and, in many existing smart-stick systems, fall detection relies primarily on fixed sensor thresholds, which may result in false alarms during sudden but normal movements [2] [3]. This project proposes a smart walking stick with machine-learning-based fall detection to provide real-time assistance, safety monitoring, and emergency communication. The proposed system integrates motion and environmental sensors to continuously acquire information related to the user's movement and surrounding conditions [4] [2] [5] [1]. Sensor data are processed to extract relevant motion characteristics and supplied to a machine-learning model trained to distinguish falls from normal activities such as walking, standing, sitting, and other daily movements [6] [7]. In addition to local assistance and obstacle awareness, the system incorporates wireless connectivity and a mobile application to communicate detected fall events to designated caregivers, family members, or medical personnel [4] [2] [3]. The proposed approach aims to improve fall-detection reliability while reducing false alarms compared with simple threshold-based techniques. By combining embedded sensing, machine learning, and IoT-enabled emergency notification within a single assistive device, the system provides a practical and intelligent solution for improving the safety, independence, and remote monitoring of vulnerable users.**

***Index Terms*—IEEE, IEEEtran, journal, LaTeX, paper, template.**

I. Introduction

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August 26, 2015

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II. Conclusion

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Appendix A
Proof of the First Zonklar Equation

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Appendix B

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Acknowledgment

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